a
VaxPress
5′UTR AUG CDS UAA 3′UTR A100
MFE ↓
CSI ↑
−MFE ↑
Orthrus TE ↑
CSI ↑
RiboDecode ↑
0.75
0.5
0.25
1
CAI ↑
Half-life ↑
Initial
VaxPress
MORSE
MORSE
5′UTR AUG CDS UAA 3′UTR A100
MFE ↓
CSI ↑
Balanced selections
100
75
50
25
CSI percentile
0.3
0.4
0.5
−MFE [kcal/(mol·nt)]
b
LinearDesign
5′UTR AUG CDS UAA 3′UTR A100
MFE ↓
CAI ↑
−MFE ↑
Orthrus TE ↑
CSI ↑
RiboDecode ↑
0.75
0.5
0.25
1
CAI ↑
Half-life ↑
MORSE
5′UTR AUG CDS UAA 3′UTR A100
MFE ↓
CAI ↑
Orthrus TE ↑
Sequence feasibility
Initial
0/117
BNT162b2
HBB
MFE specialist
8/117
RPL32 ΔTOP
PSMB3
TE specialist
26/117
HSD17B4
HBB
CAI specialist
12/117
RPL32 ΔTOP
HBB
Best HV
13/117
HSD17B4
HBB
c
RiboDecode
5′UTR AUG CDS UAA 3′UTR A100
Ribosome load ↑
−MFE ↑
Orthrus TE ↑
CSI ↑
RiboDecode ↑
0.75
0.5
0.25
1
CAI ↑
Half-life ↑
Initial
Best HV
MORSE
5′UTR AUG CDS UAA 3′UTR A100
Ribosome load ↑
CSI ↑
Sequence feasibility
Native
Best HV
1
2
3
4
5
1/5 pass
5/5 pass